

Fachhochschule Vorarlberg GmbH		Author: Horatiu O. Pilsan	
EIT & MEC - Digi SGL 4. Topic: State Machines	Rev. 1.1	27.05.2026	Page 1 of 1

1. Objective:

Understand the mode of operation of finite state machines implemented with hardware.

2. Questions:

1. What are finite state machines?
How can finite state machines be described using state chart diagrams?
Which are the basic elements of these diagrams?
2. What is the difference between Moore and Mealy state machines?
3. Understand the example state machine from the example presentation:
one simple traffic light for cars reacting on a switch pressed by pedestrians
4. Implement the Moore state machine by filling in:
the transient functions: which state will occur after which state under which condition,
the output functions: which output will be set in which state.

4. Tasks:

1. Sum up the most important pieces of information on the topic on one flip chart without adding too many details! Best suited is a visual manner.
2. Formulate questions on the topic, which you were not able to answer, neither alone nor in the team meeting!

5. Dates:

See course schedule in the information file for the course (_EIT-Digi_Info.pdf).

6. Literature:

- [1] Fricke, Klaus (2014): Digitaltechnik- Lehr- und Übungsbuch für Elektrotechniker und Informatiker, 7. Auflage, Wiesbaden: Springer-Vieweg (Kap. 8, p. 93-104)
- [2] Witowitz, Roland; Urbanski, Klaus; Gehrke, Winfried (2012): Digitaltechnik. Ein Lehr- und Übungsbuch. 6. Auflage. Heidelberg: Springer (Kap. 6.4, p. 206-213)
- [3] Gehrke, Winfried; u.a. (2016): Digitaltechnik. Grundlagen, VHDL, FPGAs, Mikrocontroller, Berlin: Springer-Vieweg (Kap. 5.2, p. 130-144)